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Palaeomagnetic study of the (Late Precambrian) West Greenland kimberlite–lamprophyre suite: definition of the Hadrynian Track

J.D.A. Piper

Sub-Department of Geophysics, University of Liverpool, P.O. Box 147, Liverpool, L69 3BX (Great Britain)

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Kimberlite and potassic lamprophyre dykes were intensively intruded into the early Proterozoic Nagssugtoqidian mobile belt of West Greenland during an important phase of brittle reactivation in Late Precambrian–Early Cambrian times (ca. 580–570 Ma) and during at least one other minor phase. Thermal and alternating field demagnetisation studies of 52 of these dykes identify primary components residing in the critical blocking temperature range distributed between shallow westerly and steep positive directions. Near the axis of the Proterozoic shear belt the dykes (predominantly lamprophyres) have closely grouped shallow directions with a reversal; near the margins of the shear belt dykes (predominantly kimberlites) have steeper and distributed directions. The cleaned components of magnetisation appear to be single, and the distribution of directions is interpreted to record a migration of the palaeofield axis which intersecting relationships show to have been from shallow to steep. The dyke directions are grouped to define representative mean palaeopoles of 215°E 3°N (LK1, $A_{95} = 3.9°$), 213°E 18°N (LK2, $A_{95} = 6.1°$), 203°E 46°N (LK3, $A_{95} = 10.4°$) and 259°E 54°N (LK4, $A_{95} = 11.0°$); a subsidiary direction recorded in five dykes near the southern margin of the shear belt (LK5, palaeopole at 297°E 16°S ($A_{95} = 12.5°$)) is derived entirely from lamprophyres and is possibly Silurian in age. An Rb–Sr isochron on three lamprophyres of 1227 Ma and agreement of the remanence direction with ca. 1220 Ma rocks from elsewhere in Greenland suggests that the LK1 component is wholly or partly of that age.

The remaining sequence of palaeopoles falls along the Hadrynian Polar Track and the age data relating to this track are re-evaluated. Evidence for a pre-800 Ma age is no longer valid and the new data from West Greenland confirm that the track is latest Precambrian to Early Cambrian in age. It is shown to connect poles of Late Precambrian and Lower Cambrian age and to embrace other data from the Laurentian shield. The rapid passage of the shield across the South Pole is consistent with the sedimentation sequences, and suggests a high-latitude origin for the tillite horizon of this age. The Hadrynian Track is also compared with the contemporaneous record from Gondwanaland and it is shown that the two shields were in juxtaposition in the identical reconstruction to the Proterozoic Supercontinent until earliest Cambrian times. This discovery links the Lower Cambrian marine transgression and the widespread ca. 580–560 Ma alkaline province in the Gondwanaland, Laurentian and Fennoscandian shields to major continental break up, and it conforms with evidence that the Iapetus Ocean did not open until Cambrian times.

1. Introduction

Geochronological studies summarised by Doig (1970), MacIntyre (1970) and Kresten et al. (1977) demonstrate that the majority of the minor alkaline bodies associated with the St. Lawrence graben, the Labrador and Greenland coasts, the Oslo graben and the Baltic coast represent a single igneous province of latest Precambrian/earliest Cambrian age. This ca. 565 Ma event (Doig, 1970)

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presents an important opportunity to determine the palaeomagnetic field at this time, which is itself of special interest because studies elsewhere (Hagstrum et al., 1974; McElhinny et al., 1974) show that the Late Precambrian/Early Cambrian interval was a period of exceptionally rapid apparent polar wander (a.p.w.) movement relative to at least some of the continents.

2. Outline geology and age

In western Greenland, kimberlite and lamprophyre dykes are closely associated with the Early Proterozoic Nagssugtoqidian mobile belt between Holsteinsborg and Itivdleq and are largely confined to the Ikertôq shear belt (Fig. 1). As elsewhere in this North Atlantic province, intrusion has followed pre-existing lines of weakness: outside of this zone in Greenland alkaline intrusions are apparently rare and are mostly of different age (Larson and Møller, 1968; Doig, 1970) although the continuation of the Nagssugtoqidian belt in eastern Greenland contains a few isolated kimberlite dykes apparently of the same age as the West Greenland province (Mitchell, 1978). The Nagssugtoqidian belt is a zone of Archaean amphibolite and granulite facies gneisses strongly reworked by ductile deformation during Early Proterozoic times: a strong ENE tectonic fabric was imposed by a (Nag 1) deformation prior to intrusion of the Kangamiut dykes at 1950 ± 60 Ma (Kaalsbeek et al., 1978); a later episode of ductile deformation (Nag 2) folded these dykes and the Nag 1 fabrics (Watterson, 1974) and culminated in ENE trending zones of overthrusting to the SSE. Later brittle fracture is represented by pseudotachylite veins throughout the Ikertôq shear belt but concentrated near its northern and southern margins. Kimberlite dykes were first reported in this zone by Escher et al. (1970) and subsequently described by Escher and Watterson (1973). Some thicker bodies contain nodules, but they are mostly thin (0.5–2 m) dykes up to 1–2 km in strike length with a strike direction (100–120°E) strongly controlled by basement structures. Associated lamprophyre dykes are generally less than 1 m thick with strike directions of 75–90°E, and comprise mica, olivine and pseudoleucite in a carbonate-rich matrix. Both types of dykes have chilled margins and are frequently banded as a result of multiple or pulsatory injection. Petrologic and geochemical studies of the West Greenland alkaline rocks by Scott (1977) suggest that the kimberlite magma originated at depths greater than 170 km and temperatures above 1100°C. A close correlation between basement structure and alkaline rock type is also apparent: the potassic lamprophyres are confined to the (Ikertôq and Itivdleq) shear zones and are mostly intruded into amphibolite facies rocks (Scott, 1977 and Fig. 1) while kimberlite dykes intrude predominantly granulite facies rocks near to, but predominantly outside of, the shear belts. The petrological variety is considerable because the province probably also includes the Sarfartôq carbonatite complex (Secher and Larson, 1978) intruded near the Nagssugtoqidian boundary and cut by radiating carbonatite dykes and breccias.

Bridgwater (1971) reported a K–Ar whole rock age of 594 ± 18 Ma (age adjusted to revised decay constant (Dalrymple, 1979)) for a kimberlite 100 km east of Itivdleq near the southern margin of the zone. Similar ages in the range 570–580 Ma were determined by McIntyre et al. (1980) using K–Ar (phlogopite) and Rb–Sr studies on a kimberlite from the northern margin of the zone (site 3 of this study). These determinations compare with ages falling between 569 and 579 Ma from the St. Lawrence graben (Doig, 1970), an age of 610 Ma for a carbonatite at Umanak 500 km north of Holsteinsborg (Larson and Møller, 1968), and an age of 553 ± 6 Ma from a kimberlite dyke trending 100°E in the Nagssugtoqidian belt in eastern Greenland (Mitchell, 1978). Recent Rb–Sr studies of G.B. Smith (data given by Scott, 1981) on three kimberlite dykes yield an isochron of 587 ± 24 Ma but also suggest that some of the lamprophyres are appreciably older: an isochron based on two dykes from Sagdlerssuaq and one from Itivdleq yielded an age of 1227 ± 12 Ma.

The total time interval represented by the alkaline bodies studied here can only be estimated but it was probably appreciable because: (1) the magmas are mantle derived (Scott, 1977) and sometimes include mantle xenolithic material; and (2) intrusion of the kimberlites was associated with

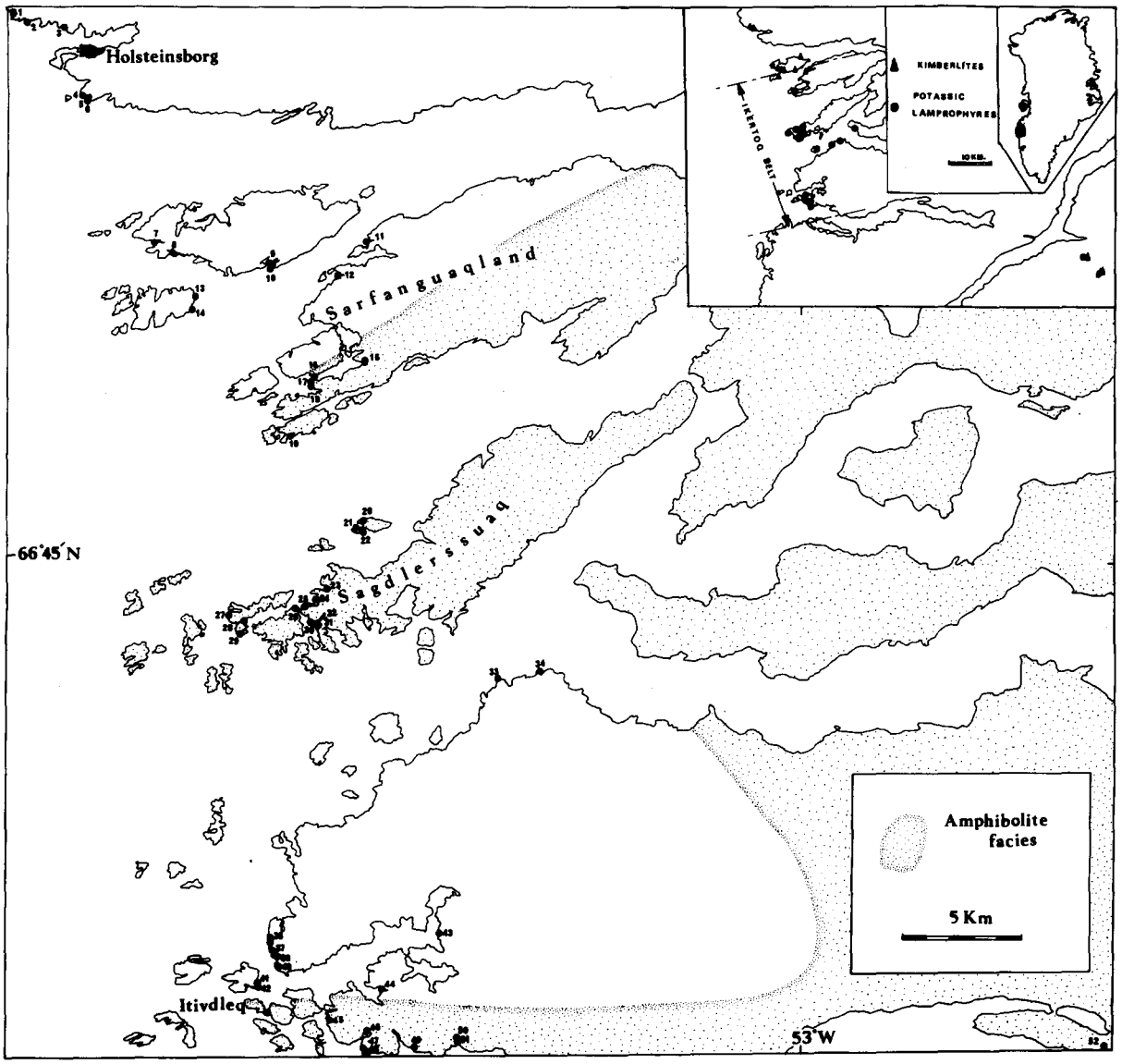


Fig. 1. Palaeomagnetic sampling sites in kimberlite and lamprophyre dykes intruding the basement terrain of central western Greenland. The inset diagram shows the distribution of petrologic types according to Scott (1977). Note the association of the lamprophyres with the amphibolite-facies terrain and the axial part of the Lower Proterozoic shear belt in this area (see also Fig. 6). Blank areas are granulite facies terrain.

brittle fracture events represented by pseudotachylite veins which in part postdate, but mostly pre-date, the alkaline intrusions (J. Watterson, 1975 and personal communication). One pseudotachylite contains primary glass; collectively they yield a coherent remanence direction including one field reversal (Piper, 1981a).

3. Field and laboratory methods

Cores were drilled in 52 dykes within this kimberlite–lamprophyre suite using a portable petrol-driven motor at the sites illustrated in Fig. 1; they comprise the bulk of the known kimberlite and lamprophyre localities in this zone. At least seven, and sometimes as many as 25, separate cores were oriented from each unit by sights on the Sun or by magnetic bearings in conjunction with topographic sites. In the laboratory the cores were cut into 2.4 cm cylinders and measured before and after progressive alternating field (AF) 500 c/s treatment with peak fields incremented in small steps, generally of 5 milliTesla (mT). In addition, at least one core from each site was progressively thermally demagnetised using a modified Schonsted TSD-1 demagnetiser at temperatures of 150, 300, 400, 500°C and smaller steps of 20 or 30°C to the Curie points of magnetite (580°C) or hematite (680°C). Magnetic moments were measured with an automated parastatic magnetic capable of accurately reproducible readings down to intensities of 0.02 A m^2 kg^{-1}; a small number of cores, mainly from the potassic lamprophyres, were too weak to measure. To facilitate identification of magnetic phases, thermomagnetic determinations (saturation magnetisation, J_s versus temperature) were made on at least one core from each site.

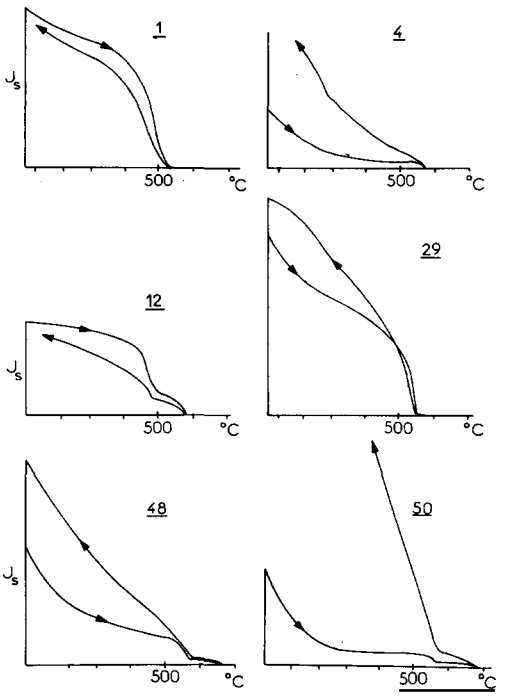


Fig. 2. Some representative examples of the different types of thermomagnetic determinations (saturation magnetisation J_s versus temperature) derived from the kimberlite and lamprophyre dykes; the J_s axis is arbitrary.

4. Palaeomagnetic results

The thermomagnetic determinations (Fig. 2 and Table I) identify titanomagnetite with Curie points between 480 and 580°C as the magnetic constituent at most sites; in two kimberlite dykes (12, 13) in the north two distinct phases with Curie points of about 490 and 580°C are present. In four dykes near the southern margin of the Ikertôq shear belt (48, 49, 50 and 52) hematite is also identified as a magnetic constituent by both thermomagnetic analysis (Fig. 2) and thermal demagnetisation, and by the latter technique at a few other sites, notably number 30 (see Fig. 4). A lower Curie point at about 300°C is present in a number of sites in the north and may be due to pyrrhotite (visible sulphides are present in many of the more coarse-grained samples) but is generally only clearly visible in the cooling curves (e.g. 4, Fig. 2).

Behaviour to AF cleaning is fairly consistent within sites but variable between sites: some have low coercivities and exhibit a rapid fall in remanence (e.g. 8, 12, 13, Fig. 3); median destructive fields are correspondingly low and NRM's show erratic behaviour at higher fields. Some of the kimberlites have strong total NRM's but exhibit only poorly-stable behaviour and erratic changes in intensity with AF treatment; these samples are rejected and the site group means (Table I) are accordingly based on a smaller number of samples than the total number of cores collected. In general, however, decrease in moment is practically systematic in fields below 50 mT although some hematite-bearing samples exhibit little change. Directional movement with this decrease frequently exhibits the progressive subtraction of a steep low coercivity component possibly related to the pre-

TABLE I

Palaeomagnetic results, western Greenland kimberlite–lamprophyre suite

Site No.	Rock type	J_o ($\times 10^{-5}$ A m^2 kg^{-1})	N	D	I	R	k	Curie point(s)	Curve type
Holsteinsborg area									
1	K	5.132–1.452	8	295.7	72.6	7.82	40	525	A
2	K	4.006–1.906	7	288.6	23.0	6.34	9	550	A
3	K	3.873–1.340	(i)8	106.0	64.0	7.86	49	520	B
			(ii)5	250.9	29.1	4.82	22		
4	K	0.010–0.002	4	345.8	29.8	3.95	57	300, 570	C
5	K	0.013–0.002	5	275.2	28.4	4.86	29	290, 565	C
6	K	0.234–0.007	10	263.4	58.5	9.53	19	290, 570	C
Manitorssuaq–Sarfangaqualand									
7	K *	2.454–0.680	7	283.0	37.6	6.60	15	560	B
8	K	4.729–0.031	13	274.3	6.8	10.97	6	—	—
9	K	0.007–0.002	6	293.2	38.0	5.84	31	300, 580	C
10	K	0.038–0.005	6	267.6	10.3	5.83	30	570	C
11	K *	1.583–0.507	7	267.5	80.5	6.42	10	480	B
12	K	2.478–0.002	7	259.1	4.2	6.69	20	490, 580	B
13	K	1.522–0.320	8	246.3	77.3	7.79	34	500, 580	B
14	K	1.304–0.426	6	223.6	82.3	5.79	24	575	A
15	K	0.098–0.002	8	261.6	30.1	7.45	13	570	A
16	K	0.199–0.006	7	276.2	24.7	6.86	43	580	C
17	K	0.012–0.002	6	269.9	3.7	5.70	17	570	D
18	K	0.040–0.006	7	294.2	72.7	6.95	132	570	A
19	L	0.016–0.001	6	260.9	23.9	5.66	15	570	A
Sagdlerssuaq									
20	L	0.007–0.003	7	323.4	53.4	6.69	19	570	D
21	L	0.038–0.005	7	277.9	−3.7	6.82	33	560	D
22	L	0.561–0.003	7	265.9	−2.5	6.85	41	570	D
23	K	5.322–2.436	8	280.8	73.1	7.74	27	545	B
24	K	0.008–0.005	7	280.3	−11.6	6.80	30	545	C
25	K	0.005–0.002	5	—	—	—	—	570	D
26	L	0.009–0.003	8	202.9	−5.2	7.38	11	570	D
27	—	0.021–0.005	7	265.7	8.8	6.74	23	580	C
28	L	0.042–0.004	7	118.0	64.5	6.98	274	550	C
29	—	0.015–0.006	6	260.3	13.3	5.81	27	550	C
30	L	0.056–0.009	8	271.8	−5.4	7.96	160	570	C
31	—	0.028–0.011	7	287.9	22.8	6.94	101	550	C
32	—	0.870–0.661	8	274.1	3.4	7.53	15	570	B
33	L	32.425–0.035	8	261.8	68.6	7.71	24	560	A
34	L	3.449–0.403	6	271.1	77.6	5.93	76	565	A
Itivdleq Fjord and district									
35	—	0.018–0.004	7	295.7	−0.4	5.95	6	580	D
36	—	0.006–0.002	8	285.3	17.9	7.58	17	565	C
37	—	0.016–0.002	4	293.4	62.7	3.96	62	570	C
38	—	0.011–0.003	8	187.8	−9.6	7.42	12	565	C
39	—	0.017–0.002	7	192.5	6.9	6.61	16	565	C
40	—	0.007–0.003	4	244.2	36.7	3.82	16	570	C
41	—	0.014–0.003	7	192.6	18.6	6.30	9	560	D
42	—	0.036–0.011	7	197.3	31.3	6.56	14	570	C
43	—	0.024–0.003	8	287.3	17.5	7.74	28	570	D
44	—	0.005–0.002	7	101.6	−4.2	6.76	26	565	C
45	L	6.892–0.886	5	184.1	61.0	4.89	36	550	C

TABLE I (continued)

Site No.	Rock type	J_o ($\times 10^{-5}$ A m^2 kg^{-1})	N	D	I	R	k	Curie point(s)	Curve type
46	L	0.020−0.002	8	305.7	48.8	7.80	36	565	C
47	L	0.031−0.003	8	295.7	66.0	7.69	23	570	C
48	L	0.029−0.007	7	277.8	73.3	6.94	96	570, 670	C
49	L	0.186−0.008	7	274.2	12.6	6.67	18	570, 680	C
50	L	0.014−0.004	8	176.9	17.1	7.73	26	570, 680	D
51	L	0.016−0.003	5	298.4	24.2	4.99	285	565	C
52	L	9.135−4.258	6	304.6	53.3	5.78	22	540, 670	C

* Contains ultramafic nodules.

K = kimberlite; L = lamprophyre.

D and I are the mean declination and inclination of N samples with resultant vector R and precision parameter k $[(N-1)/(N-R)]$. Thermomagnetic curves illustrate similar heating and cooling curves (A); cooling curves below heating curves (B); cooling curves above heating curves (C), or a very large increase in saturation magnetisation on cooling (D).

sent field (except at site 3 where a distinct low coercivity component ((i) in Table I) is present in a number of cores), followed by no systematic change (e.g. 8, 12, 16, 17, 29, Fig. 3). Other samples show no significant migration and retain the steep, intermediate or shallow inclinations exhibited by the total NRM although the intensity is progressively reduced by AF treatment (e.g. 1, 18, 20, 27, 51, Fig. 3). That these end points isolated by the AF demagnetisation are representative of the primary remanence in these rocks is, in general, confirmed by thermal demagnetisation (Fig. 4). With this technique many of the cores also exhibit the progressive subtraction of a steep positive component (e.g. 13, 19, 27, Fig. 4) over the broad range of treatment 150–500°C comparing with similar movement over the limited range of 5–25 mT, or less, of AF treatment. Other sites however (e.g. 30, 45, 47, 48, Fig. 4) retain the shallow intermediate or steep inclinations of the total NRM's to the critical blocking temperature ranges, and confirm that these are the only components present. In many samples close to the Curie point of magnetite, more magnetite is produced after sulphides or silicates by the thermal demagnetisation procedure this produces an upturn in the J/J_0 curves (Fig. 4) and directions disperse or are non-systematic about a mean point.

Site mean directions are listed in Table I and are calculated using a least-scatter criteria similar to that of Dagley and Ade-Hall (1970) with the range of utilised fields selected by inspection of the AF and thermal demagnetisation plots; total NRM components are frequently dominated by the soft steep positive components and are not listed.

The collective site mean directions are plotted in Fig. 5. They are clearly non-Fisherian and are distributed from shallow to steep westerly directions with a minor group of southerly directions. A purely bimodal distribution might be expected from the observation that these rocks appear to be of at least two different ages namely ca. 570 Ma and ca. 1230 Ma but this does not immediately explain this distribution of directions. Indeed even a mixture of these two palaeofield directions is not applicable because the palaeofield directions of the Laurentian Shield at ca. 1230 and 600 Ma appear to have been closely similar (Morris and Roy, 1977; Piper and Stearn, 1977). This distribution might be interpreted as either: (a) migration of the palaeofield direction with respect to the sample area; or (b) to 'smearing' between at least two palaeofield directions, notably a primary and a secondary components, with the total remanences being the vector sums of these two components in different ratios. The former explanation is favoured for the following six reasons:

(1) These rocks are unaltered, come from fresh

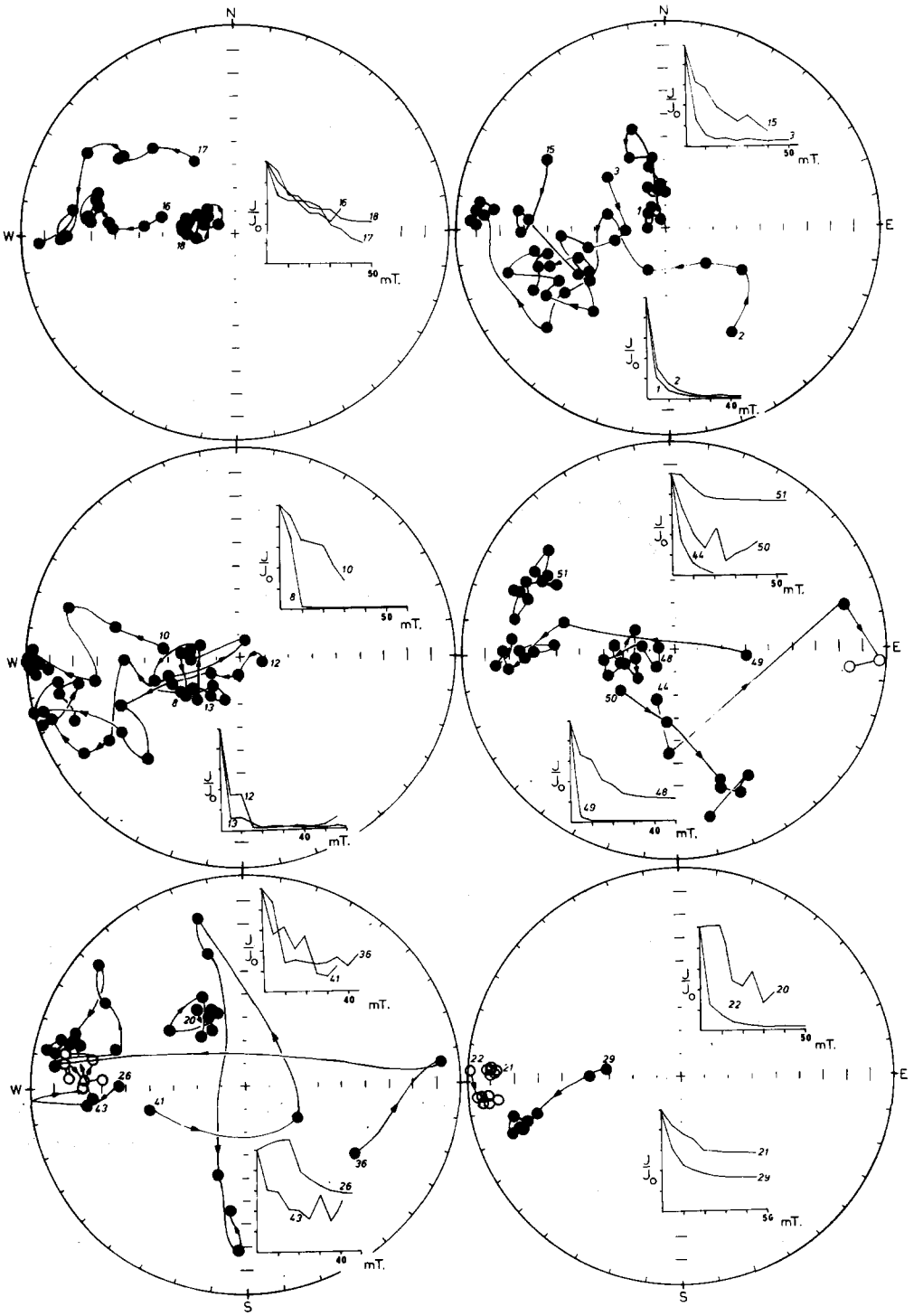


Fig. 3. Equal area projections showing behaviour of directions of magnetisation and magnetic intensities (expressed as a fraction of the initial intensity, J_0) with progressive alternating field demagnetisation. Closed symbols are lower hemisphere and open symbols are upper hemisphere plots.

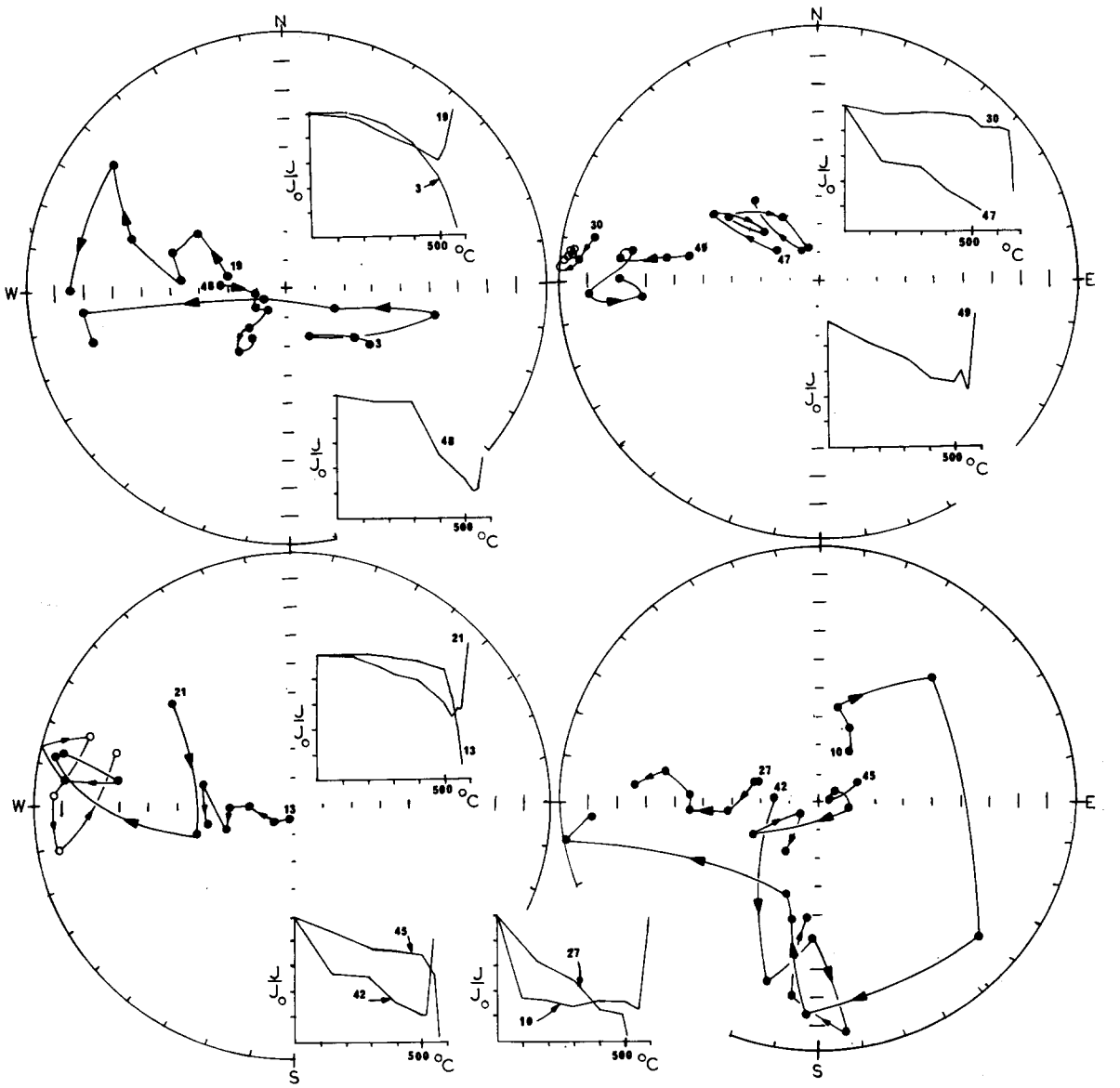


Fig. 4. Examples of the behaviour of the directions of magnetisation and magnetic intensities in the kimberlite and lamprophyre dykes with progressive thermal demagnetisation. Symbols are as for Fig. 3.

highly-glaciated terrain, and no post-Cambrian events which could have achieved a widespread but patchy remagnetisation are recognised.

(2) The end points identified by AF demagnetisation are confirmed by thermal demagnetisation in the critical blocking temperature range, and they are predominantly magnetite-held.

(3) There is no clear relationship of sampled properties to the magnetisation directions, and low coercivity samples have both high and low inclinations, as do both strongly- and weakly-magnetised sites; the steep positive directions are not held by high blocking temperature hematite-components ineffectively subtracted by the demagnetisation techniques.

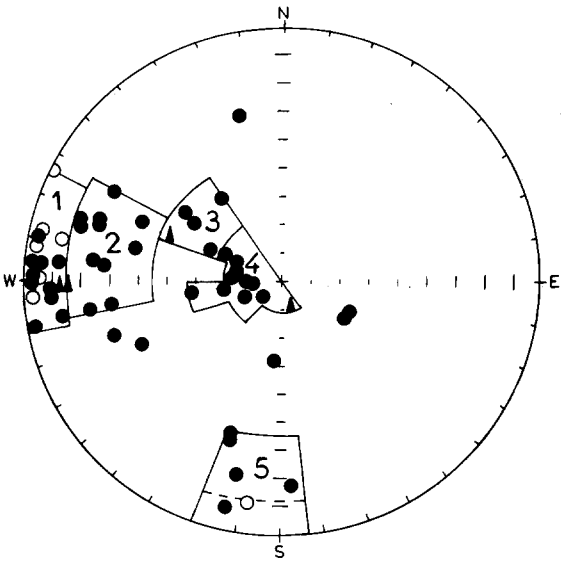


Fig. 5. The distribution of cleaned site mean directions from the kimberlite and lamprophyre dykes derived from this study plus directions from four sites (triangles) reported by Beckmann and Mitchell (1976). The groups 1–5 are those used to calculate the representative group mean directions (Table II).

(4) Two observations show that intrusion of the province spanned a considerable interval of time. Firstly, the rocks are of mantle origin (Scott, 1977) and secondly, intersecting relationships show that some dykes predate pseudotachylite veins (the latter representing brittle fracture events) while most post-date them. Palaeomagnetic study of the pseudotachylites defines a strong stable magnetite-held remanence of dual polarity $D = 269°$, $I = 22°$ (Piper, 1981a) which is comparable to many of the kimberlite and lamprophyre site means. That this shallow inclination direction, at least, is an ancient palaeofield axis is confirmed by the reversal at site 44 (Table I).

(5) When the site mean directions are plotted area by area, they are clearly systematically related to their position within the Lower Proterozoic shear belt (Fig. 6). At Sagdlerssuaq near the axis of the belt, the bulk of the sites are closely grouped with westerly shallow negative to positive directions of magnetisation. At Manitorssuaq and Sarfanguaqland somewhat further removed from the axis, directions are still westerly, but steeper positive. Pseudotachylites are abundant here and predate kimberlite sites 13 and 18; they thus define the direction of migration of the ancient palaeofield axis. At the northern and southern margins of the shear belt directions of magnetisation are more scattered and generally of steeper inclination. Scott (1977) has shown that rocks near the axis of the shear belt are predominantly potassic lamprophyres intruded into amphibolite facies terrain; near the margins they are mainly kimberlites and intruded into granulite facies terrain.

(6) It is conceivable that the cleaned steep directions are composite and the resultant of a Late Precambrian and a Cretaceous component acquired during rifting and opening of the Labrador Sea, and with blocking temperature spectra so similar that they have not been separated by the cleaning treatment. This again is not supported by the distribution of directions which are predominantly shallow near the axis of the reactivated zone (which includes pseudotachylites and a local retrogression of mineral phases) and predominantly steep in the stable areas outside of this zone. The kimberlite–pseudotachylite province represents the reactivation of this ancient Proterozoic shear belt at ca. 580–560 Ma. Some, but not all, of the lamprophyres may be of this age. The palaeofield direction for Greenland at 1230 Ma is accurately defined from the Gardar Province of South Greenland at $D = 292°$, $I = -15°$ (Piper and Stearn, 1977) and this would accord with the westerly negative directions observed in some lamprophyre dykes from the axial parts of the shear belt. (Unfortunately, the dykes used for the lamprophyre isochron of Scott (1981) do not correspond to the palaeomagnetic sample.) The southerly directions of magnetisation represented in some lamprophyre dykes near the southern extremity of the belt may be significantly younger in age. In post-Late Precambrian times they correlate only with certain Silurian field directions (Section 5 and Fig. 7) from North America; the palaeomagnetic data predict that they may be of this age and Doig and Barton (1968) find, in addition to the main ca. 560 Ma event, that a subsidiary alkaline event at 420–460 Ma is present in the St. Lawrence rift.

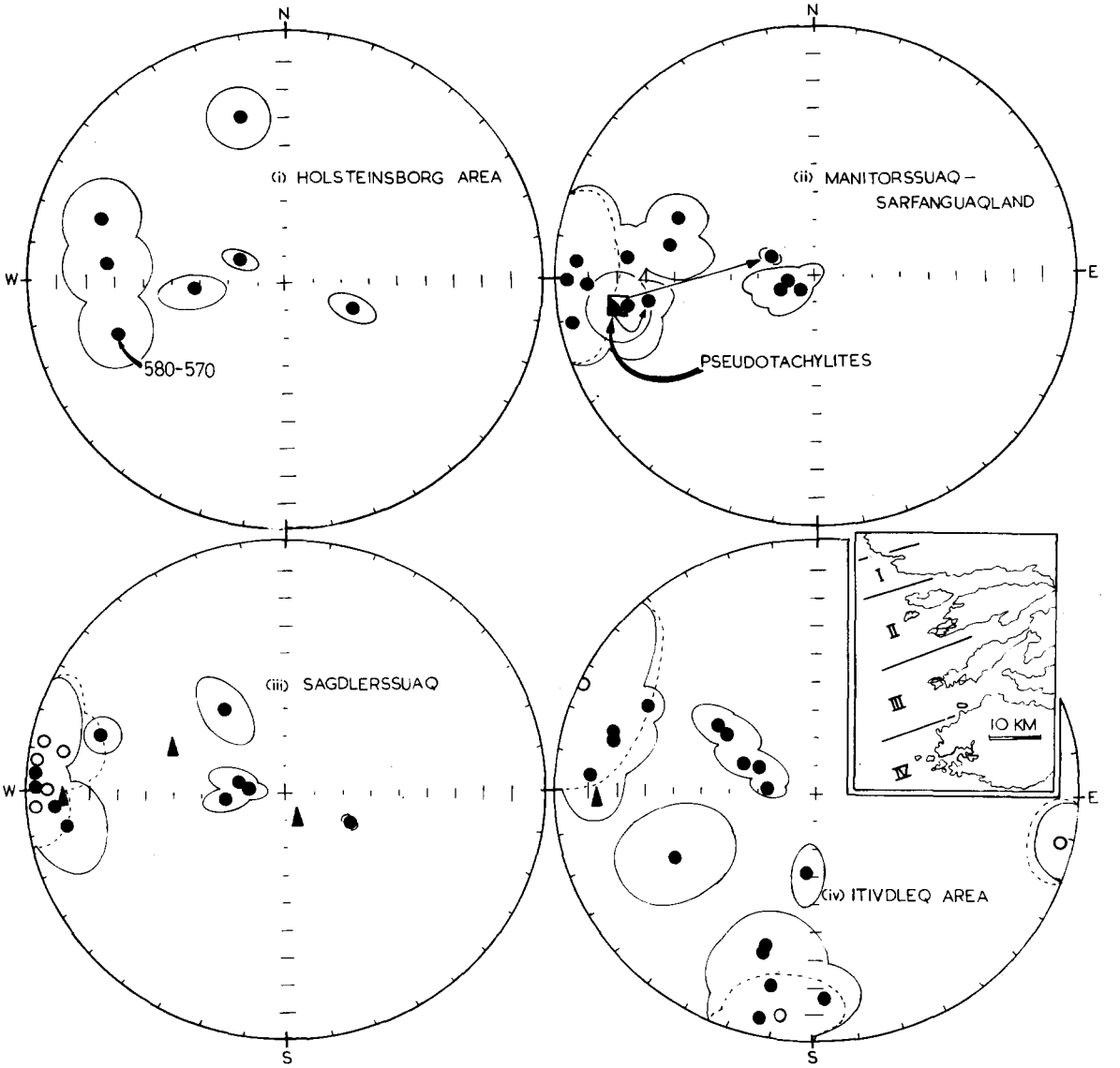


Fig. 6. The site mean directions of magnetisation from the kimberlite and lamprophyre directions of magnetisation from the kimberlite and lamprophyre dykes plotted area by area across the Nagssugtoqidian belt together with 95% confidence circles. The approximate extent of the Ikertôq shear belt in this area is shown in the inset diagram. The intersecting relationship of the dykes with pseudotachylite veins in the Sarfanguaqland area define the directions of palaeofield migration (arrowed) in this area.

For reasons 1–6 above the distribution of palaeofield directions in Fig. 5 is interpreted to be dominated by a migration of the palaeofield axis from shallow westerly negative to steep positive at ca. 585–565 Ma with a possible continuation to southerly positive although this latter step is not well represented by these data. To calculate representative palaeomagnetic poles for this palaeofield behaviour the sites are grouped into approximately $30 \times 30°$ sectors of the stereonet as illustrated in Fig. 5 and the group means are listed and Table II. Five mean axes of magnetisation are defined in

TABLE II

Group mean palaeomagnetic results, West Greenland kimberlite–lamprophyre suite

Pole reference	Sites included in mean calculation	N	D	I	R	k	α_{95}	Palaeomagnetic pole			
								°N	°E	dp	dm
LK1	8, 10, 12, 17, 21, 22, 24, 26, 27, 29, 30, 32, 35, 44, 49, K1, K2	17	272.8	3.9	16.66	47.1	5.3	2.9	214.8	2.6	5.3
LK2	2, 5, 7, 9, 15, 16, 19, 31, 36, 43, 51	11	281.7	26.6	10.75	40.7	7.2	17.5	211.6	4.3	7.9
LK3	20, 38, 46, 47, 52, K2	6	302.8	55.8	5.92	64.1	8.4	45.8	202.7	8.7	12.1
LK4	6, 11, 13, 14, 18, 23, 33, 34, 48, K2	10	264.2	76.2	9.86	62.1	6.2	53.7	258.6	10.6	11.4
LK5	38, 39, 41, 42, 50	5	189.3	13.0	4.83	23.0	16.3	−16.4	297.0	8.5	16.6

The site means are subdivided as shown in Fig. 5 to compute these group means.
dp and dm are the semi-axes of the oval of confidence ($P=0.05$) about the mean palaeomagnetic pole in the colatitude direction and at right angles to it respectively, and α_{95} is the radius of the circle of confidence about the mean direction. Other symbols are as for Table I.
Sites with the prefix "K" are listed by Beckmann and Mitchell (1976).

this way and yield the mean palaeopoles in age sequences LK1–LK5 (Table II) with the reservation that the shallow westerly direction (LK1) probably incorporates a mixture of Late Precambrian and ca. 1230 Ma material. Unfortunately, since the palaeofield directions at ca. 1200 and ca. 600 Ma were similar there is no way in which the older material can be isolated on palaeomagnetic grounds alone.

5. Interpretation of the palaeomagnetic poles: the Hadrynian Loop

The observation that the palaeomagnetic field underwent a rapid movement with respect to North America in Late Precambrian to Early Cambrian times as outlined by the poles LK1–LK5 is not unexpected. Rapid a.p.w. movements of this general age have been identified from Gondwanaland (McElhinny and Embleton, 1976; Piper, 1976) and several of the pre-Ordovician massifs incorporated in the Armorican orogenic belt (Hagstrum et al., 1980; Piper 1981b). Indeed, Morris and Roy (1977) have already outlined an a.p.w. track known as the Hadrynian Track) which is nearly identical to that defined by poles LK1–LK5. Unfortunately, this track has hitherto been based on poorly-dated secondary and tertiary components of magnetisation in the multicomponent magnetisation record of certain sediments and doubt has been cast on its validity (e.g. Berger et al., 1979). Morris and Roy (1977) originally utilised the Lower and Upper Torridonian poles referred to Rb–Sr isochrons on shale horizons of 968 and 777 Ma to date the earlier and latter parts of their track. It has since been shown (Smith et al., 1981) that the stratigraphic sequences of poles in the Lower and Upper Torridonian are in the opposite direction to the trends required by Morris and Roy's interpretation; furthermore, the sequences and positions of these poles coincide precisely with the Laurentian a.p.w. path at 1100 and 1040 Ma, respectively, implying that the Torridonian Rb–Sr isochrons date later dewatering events (see Bath, 1974; Perry and Turekian, 1974) and not depositional (and in this instance magnetisation) events.

The evidence used to infer a pre-800 Ma Hadrynian Track is therefore no longer valid, but even if the Torridonian and late-Grenville poles (Berger et al., 1979) are excluded a number of steep inclination magnetisations of probable Late

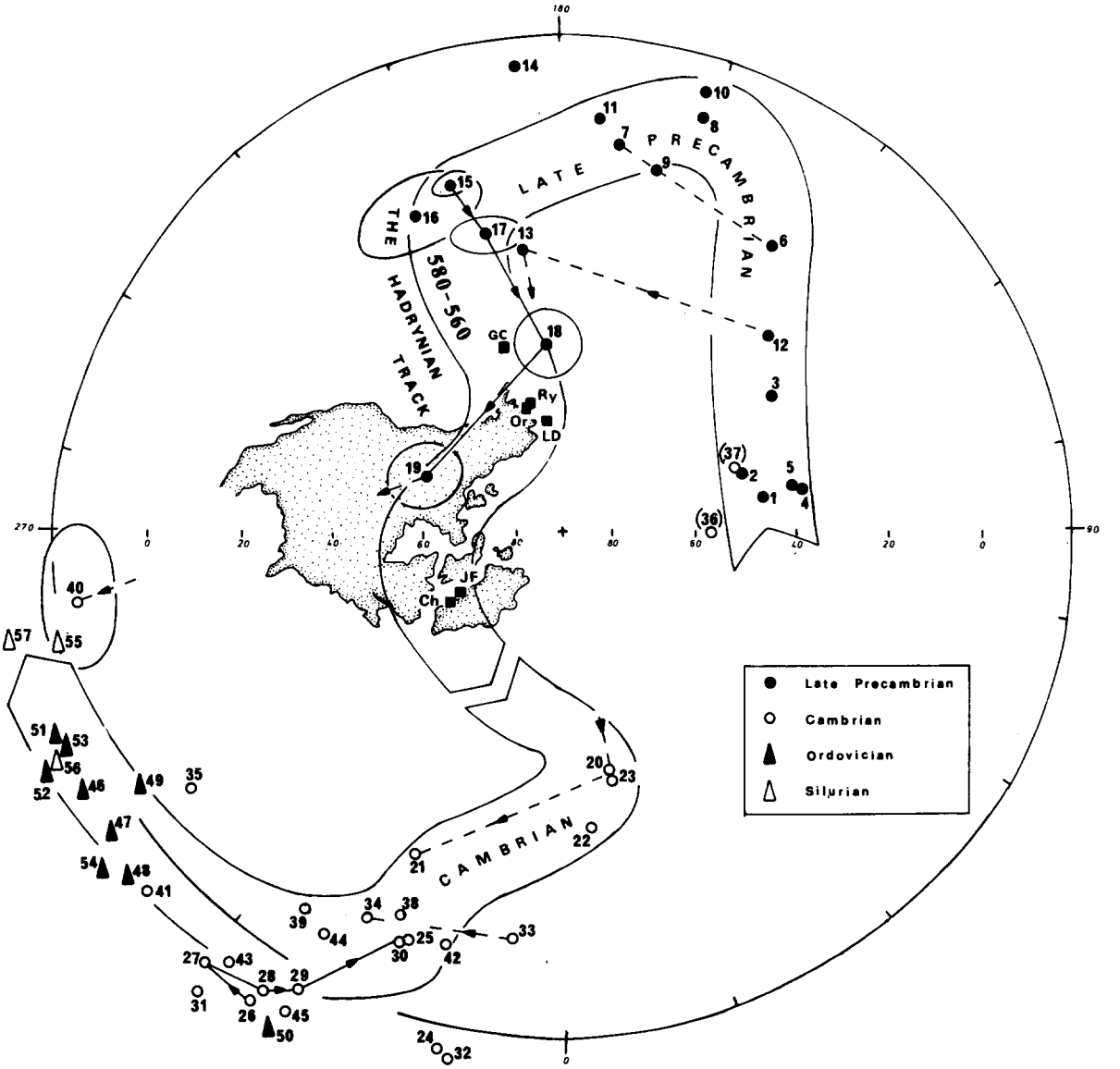


Fig. 7. The distribution of palaeomagnetic poles from the Laurentian shield assigned to the interval ca. 700 Ma to Silurian times (Table III) with a generalised a.p.w. path incorporating the Hadrynian Track and secondary magnetisations (squares) assigned to this track (Morris and Roy, 1977). The known or possible time sequences of palaeopoles are indicated by the arrowed lines (see text).

Precambrian age remain. They could be accommodated by a track incorporating the sequence of Greenlandic poles and linking other North American Late Precambrian poles ca. 600 Ma in age and Lower Cambrian poles as shown in Fig. 7 and Table III. This assessment includes the ca. 700 Ma to Silurian palaeomagnetic poles from the Laurentian shield and its margins including some Ordovician to Devonian data from the Maritime Provinces although it is possible that this zone is allocthonous with respect to the remainder of the continent (Kent and Opdyke, 1978). McElhinny et

TABLE III

Palaeomagnetic poles for North America and Greenland assigned to the interval ca. 700 Ma—Silurian times

Pole No.	Rock unit	Palaeomagnetic South Pole		Age	Reference
		°N	°E		
1	Intrusives, Colorado [a]	44	100	Late Precambrian –Lower Palaeozoic	Larson and Mutschler (1971)
2	Intrusives, Colorado	48	107	(704–485 Ma)	French et al. (1977)
3	Intrusives, Colorado	37	122		French et al. (1977)
4	Intrusives, Colorado	37	100	525 Ma(?)	Patterson et al. (1979)
5	Intrusives, Colorado	29	100		Patterson et al. (1979)
6	Iron Mountain–Maclure Mountain alkaline complex	15	142	Late Precambrian	French et al. (1977)
7		5	174	Cambrian	
8	Franklin rocks, A	−4	161	Late Precambrian	Palmer and Hayatsu (1975)
9	Franklin rocks, B	8	166		
10	Johnnie Formation	−10	162 [b]	Late Precambrian	Van Alstine and Gillett (1979)
11	Cloud Mountain basalt	5	172	605 ± 10 Ma	Deutsch and Rao (1977)
12	Lamotte Formation (I)	30	136	Late Precambrian (?)	Al-Khafazi and Vincenz (1971)
13	Lamotte Formation (E)	28	188		
14	Quebec Lava flows, A	−10	186	Late Precambrian, 564 Ma	Dankers and Lapointe (1980)
15	West Greenland, LK1	9	198 [c]		This paper
16	Pseudotachylites	15	206	ca. 580–560 Ma	This paper
17	LK2	23	194		This paper
18	LK3	51	184		This paper
19	LK4	58	248		This paper
20	Lamotte Sandstones, 2a	38	12	Cambrian	Al-Khafazi and Vincenz (1971)
21	Lamotte Sandstones, 2b	12	335		Al-Khafazi and Vincenz (1971)
22	Lodore Formation	23	6	Cambrian	Pole 3/76
23	Bonnaterre Dolostone	35	10	Cambrian	Beales et al. (1974)
24	Waynesboro Formation	−28	345	Late–Middle Cambrian	Watts et al. (1980b)
25	Wilberns Formation	−6	339	Upper Cambrian	
26	Hickory Sandstone	−34	325		Watts et al (1980a)
27	Cap Mountain Limestone	−33	320	Middle–Upper Cambrian	Watts et al. (1980a)

28	Lion Mountain Limestone	−27	326	pole in	Watts et al. (1980a)
29	Welge Sandstone/Morgan Creek Limestone	−24	331	stratigraphic sequence	Watts et al. (1980a)
30	Point Peak Shale	−6	339		Watts et al. (1980a)
31	Rome Formation, Tennessee	−39	322	Late–Middle Cambrian	Watts et al. (1980b)
32	Bradore Sandstones	−29	347	Lower Cambrian	Rao and Deutsch (1976)
33	Quebec Lavas, B	−1	353		Dankers and Lapointe (1980)
34	Quebec Lavas, C	−6	334		Dankers and Lapointe (1980)
35	Ratcliffe Brook Formation	−10	304	Lower Cambrian	Pole 8/148
36	Abrigo Formation, Arizona	59	89 [b]	Middle–Upper Cambrian	Elston and Bressler (1977)
37	Muav Formation, Arizona	55	110 [b]	Middle–Upper Cambrian	Elston and Bressler (1977)
38	Tapeats Sandstone, Arizona	−5	338 [b]	Lower–Middle Cambrian	Elston and Bressler (1977)
39	Ophiolite Complex, Quebec	−13	326	Middle Cambrian?	Seguin (1976)
40	(LK5), West Greenland	−15	278	?	This paper
41	Wichita Granites	−31	310		Vincenz et al. (1975)
42	Wichita Granites	−4	344		Spall (1970)
43	Wichita Granites	−30	322	Late Cambrian 525 Ma	Ku et al. (1967), recalculated by Van der Voo et al. (1976)
44	Wichita Granites	−33	327		Spall (1970)
45	Wichita Granites	−16	329		Vincenz et al. (1975)
46	St. Georges Limestone	−30	299	Early Ordovician	Beales et al. (1974)
47	St. Georges Dolostone	−29	305	Early Ordovician	Beales et al. (1974)
48	Moreton's Harbour Basalt	−32	310	Late–Middle Ordovician	Deutsche and Rao (1977)
49	Quebec dykes	3	303	497 Ma	Dankers and Lapointe (1980)
50	Mocassin and Bays Formations	−33	327	Middle Ordovician	Watts and Van der Voo (1979)
51	Chapman Ridge Formation	−27	292	Middle Ordovician	Watts and Van der Voo (1979)
52	Trenton Limestone	−36	296	Middle–Upper Ordovician	McElhinny and Opdyke (1973)
53	Juniata Formation	−32	294	Upper Ordovician	Van der Voo and French (1971)
54	Beemerville Alkaline Complex	−35	306	Upper Ordovician	Proko and Hargraves (1973)
55	Castanea Formation	−21	285	Late Silurian	McElhinny and Opdyke (1973)
56	Rose Hill Formation	−26	296	Middle Silurian	French and Van der Voo (1977)
57	Bloomsburg Formation	−32	282	Upper Silurian	Roy et al. (1967)

[a] It is not known which, if any of poles 1–7 are Late Precambrian; they may all be Lower Palaeozoic in age.

[b] No local rotation assumed, see reference.

[c] Pole positions (Table II and Piper (1981a) corrected for phanerozoic drift of Greenland (Bullard et al., 1965)); LK1 includes 1230 Ma material. Two older results (22 and 35) are referenced by their number in the listing of the Geophysical Journal of the Royal Astronomical Society.

al. (1974) first suggested that a long track might be present in the Laurentian a.p.w. path in Late Precambrian to Early Cambrian times when they interpreted the magnetisations identified by Al-Khafazi and Vincenz (1971) from the Lamotte Formation of Missouri as primary magnetisations and placed them in stratigraphic sequence. This would accord with the interpretation of Fig. 7 which separates those poles or probable or known Late Precambrian age from those which are of Cambrian age, and connects them via the sequence of poles from Greenland of known latest Precambrian to Early Cambrian age. Of the magnetisations utilised by Morris and Roy (1977) to define their original Hadrynian Track, it is still possible that remanence directions from the Grand Canyon Group (No. 13, CC in Fig. 7), Orienta (Or), Rapitan Group '*Y*' magnetisation (Ry in Fig. 7), the Jacobsville 'C' magnetisation (JF) and the Chequamegon Group (Ch) plotted in Fig. 7 fall along this track. In addition the excluded data from the Torridonian and the Grenville terrain yield low inclination directions and do not appreciably affect the form of the Hadrynian Track as defined by Morris and Roy (1977). Park (1980) considers that the 'C' magnetisation from the Little Dal Group also falls along this (Hadrynian) track. The least satisfactory part of this a.p.w. path is that prior to 600 Ma. Poles 36 and 37 are clearly Middle–Upper Cambrian in age and not relevant to that part of the path which they plot; Watts et al. (1980a) incorporate them into an a.p.w. loop. In addition it is not known which, if any, of the poles 1–7 are attributable to the Late Precambrian.

The Late Precambrian–Cambrian transition dated ca. 560 Ma (Cowie and Cribb, 1978) must lie close to the middle part of this track because poles 20 and above are of known Lower Cambrian or younger age. The a.p.w. motion represented by the Hadrynian Track was therefore executed in 30 Ma or a slightly shorter period of time and carried the Laurentian Shield to high latitudes and back. Cambrian a.p.w. motion is, in detail, more complex than the generalised swathe of Fig. 7 and includes a loop in late Middle to early Upper Cambrian times (poles 26–30, Watts et al., 1980). No clear trends can be recognised between Lower, Middle and Upper Cambrian poles and this may reflect uncertainties in the palaeontological age allocations. The Lamotte poles 20 and 21, for example, may be of Middle–Upper Cambrian age. Maximum depositional ages of the Lower Cambrian are estimated from Rb–Sr and U–Pb determinations on rocks overlain unconformably by Cambrian sediments; they range from 614 to 557 Ma (Cowie and Cribb, 1978), and age determinations on authigenic glauconites and illites (mostly K–Ar) suggest that the top of the 'Lower' Cambrian may be 530 Ma or even slightly younger in age (Cowie and Cribb, 1978).

The latitudinal extent of the widespread Late Precambrian–Early Cambrian glaciations (Harland, 1964) is still a subject of much debate and palaeomagnetic studies of the tillites of this age have revealed consistently low palaeolatitudes. Tarling (1974) reported low ($\sim 10°$) apparent palaeolatitudes from the Dalradian tillites of Scotland and Morris (1977) found a possible primary magnetisation in the Upper Rapitan Group indicating a palaeolatitude of 14°. Kröner et al. (1980) identify low palaeolatitudes from the Nama Group of Namibia, and McWilliams and McElhinny (1980) identify palaeolatitudes of 10–20° from glaciogenic sediments of the Adelaide System in Australia. Unfortunately, all of these results are derived from drab sedimentary rocks, and in no instance can the magnetisation interpreted as primary be attributed with certainty to the age of deposition. The observation (Fig. 7) that the Laurentian shield moved to high latitudes in latest Precambrian to Early Cambrian times raises some doubt about a low latitude origin for the glaciogenic sediments of this age. A tillite horizon occurs near the top of the Late Precambrian Hagen Fjord Group in Greenland and is separated from the unconformity of presumed Lower Cambrian age by dolomites. A tillite of latest Precambrian age is also identified in Peary Land in North Greenland and is again overlain firstly by limestone and then by dolomites. The succession is consistent with a migration of Greenland from low to high back to low latitudes at about the Late Precambrian–Cambrian boundary. The North American tillites occurring extensively in the Cordilleras are ca. 800 Ma in age and not relevant to the a.p.w. path

defined in Fig. 7 but locations relevant to the latest Precambrian–Early Cambrian interval occur in the Avalon Peninsula of Newfoundland and the Appalachians (see summary by Williams (1975)). The steep *Y* magnetisation identified by Morris (1977) from the (1100–570 Ma) Rapitan Group appears to be associated with the pigment and Morris attributes it to a diagenetic overprint; it is probably relevant to the definition of the a.p.w. path but not to the latitudinal origin of these glaciogenic deposits.

6. Tectonic implications: break up of the Proterozoic supercontinent

The recognition of the Hadrynian Track as a Late Precambrian–Early Cambrian a.p.w. swathe requires some modification of published palaeogeographic maps; these have hitherto been constructed in the assumption that North America was essentially stationary during Cambrian times (poles 21 and 24–36, Fig. 7) while Gondawaland underwent a rapid movement form high to low palaeolatitudes and back (McElhinny and Embleton, 1976; Piper, 1976). Now that a comparable palaeolatitudinal movement of the Laurentian shield is evident this conclusion must be revised.

A summary of Gondwanaland palaeomagnetic poles relating to the Late Precambrian–Ordovician interval is listed in Table IV and based on McElhinny and Embleton (1976) and Piper (1976), with additional data listed by Davies et al. (1980), Klootwijk (1980), and supplementary data from the Nama Group of Namibia from Kröner et al. (1980); the poles are plotted in Fig. 8 using the same projection as Fig. 7. The large anticlockwise a.p.w. loop executed between Late Precambrian and Ordovician times was first defined by the stratigraphic sequence of poles from sediments falling within this age bracket in Australia. Its general applicability to Gondwanaland as a whole is confirmed by the sequence of African poles dated ca. 630 to 540 Ma, and Late Precambrian–Cambrian–Ordovician in stratigraphic age (Table IV and Fig. 8), plus the correspondence of more limited data from South America, India and Antarctica with the overall picture (Fig. 8). The complete motion corresponds with the general observations of Late Precambrian tillites in southern, central and western Africa, (including those of the Taoudini basin in West Africa dated 650–610 Ma (Clauer, 1976), overlain by sediments of low latitude origin and succeeded in turn by tillites of Ordovician age in West Africa (McElhinny et al., 1974). Data from the Late Precambrian (< 686 Ma) to Early Cambrian (> 553 Ma) Nama Group of Namibia outline an earlier extension to this

TABLE IV

Late Precambrian to Silurian palaeomagnetic poles from Gondwanaland

Pole No.	Rock unit and location	Palaeomagnetic South Pole *		Age
		°N	°E	
1	Dokhan Volcanics, Egypt	−36	17	~603 Ma
2	Ourzazate Volcanics, Morocco	−30	57	LPc–C
3	Upper Nama Group, N2, Namibia	5	271	<686>553 Ma
4	Arumbera Sandstone, Australia	24	294	LPc–C
5	Antrim Plateau Volcanics, Australia	26	319	LPc–C
6	Klipheuval Formation, South Africa	16	316	LPc–C
7	Adma Diorite, Egypt	33	345	616–590 Ma
8	Arumbera Todd River, Australia	−3	344	LPc–C
9	Upper Bhander Sandstone, India	4	339	LPc–C
10	Nama Group, N3, Namibia	−2	344	LPc–C
11	Bhander Sandstone, India	−15	335	LPc–C

TABLE IV (continued)

Pole No.	Rock unit and location	Palaeomagnetic South Pole *		Age
		°N	°E	
12	Ntonya ring structure, Malawi	28	345	630±24 Ma
13	Upper Rewa Sandstone, India	−11	350	LPc–C
14	Sijarira Group, Rhodesia	2	352	LPc–C
15	Pound Quartzite, Australia	−5	6	LPc
16	Pre-Nama dykes, South Africa	85	228	653±70 Ma(?)
17	Lower Buanji Series, Tanzania	87	263	LPc–C
18	Plateau Series A, Zambia	71	173	LPc–C
19	Plateau Series B, Zambia	60	25	LPc–C
20	Nama Group secondary, Namibia	62	61	<553
21	Amouslek Tuffs, Morocco	−42	70	590–530, Cl **
22	Serie Lie de Vin, Morocco	−29	63	C_1 **
23	Aroona Dam Sediments, Australia	21	20	C_1
24	Hudson Formation, Australia	72	338	C_1
25	Purple Sandstone, India	71	357	C_1
26	Hugh River Shale, Australia	68	33	C_{1-m}
27	Moroccan Lavas	53	34	C_{1-m}
28	Sabaloka ring structure, Sudan	83	339	540 Ma
29	Salt Pseudomorph Beds, India	66	13	C_m
30	Purmarnarca sediments (i), Argentina	36	295	C
31	South Tilcara sediments, Argentina	82	329	C
32	North Tilcara sediments, Argentina	80	350	C
33	Purmarnara sediments (ii), Argentina	45	80	C
34	Ben Azzer Volcanic sediments, Egypt	47	42	532±18 Ma
35	QS Dykes, Egypt	87	304	527–480 Ma
36	Abra de Cajas sediments, Argentina	41	66	C
37	Dokhan Volcanics, Egypt	54	327	ca. 500 Ma
38	UR Dykes, Egypt	86	185	464–497 Ma
39	Lake Frome Group, Australia	43	8	C_m–O_1
40	Doornport Formation, Namibia	22	45	500–550
41	Dundas Group, Australia	31	358	Cu
42	Adras de Mauritania CO_{10}, Mauretania	55	30	C–O
43	Adras de Mauritania CO_8, Mauretania	36	52	C–O
44	Tassili sediments, Algeria	53	26	C–O
45	Mirnyy Charnockites, Antarctica	49	14	C–O, 520±24 Ma
46	Jinduckin Formation, Australia	44	10	O_1
47	Stairway Sandstone, Australia	51	49	O_m
48	Table Mountain Series, South Africa	50	349	O
49	Tumblegooda Sandstone, Australia	27	18	O
50	Salta and Jujuy redbeds, Argentina	22	358	O
52	Sør Rondane intrusions, Antarctica	18	15	485±25 Ma
	Australian Cambrian poles from Klootwijk (1980):			
53	Basal Hawker Group	−27	2	Earliest C
54	Billy Creek-Aroona Creek Formation	−37	20	C_m
55	Basal Lake Froome Group	−29	26	C_m
56	Pantapinna Formation	−36	29	C_m
57	Kangaroo Island, primary	−34	15	Earliest C

* Pole position given relative to Africa using the Smith and Hallam (1970) reconstruction of Gondwanaland.

** These two poles may be Hercynian remagnetisations and do not agree with the other poles of this age (Fig. 7).

LPc = Late Precambrian, C = Cambrian, O = Ordovician, l, m, n = lower, middle, upper divisions.

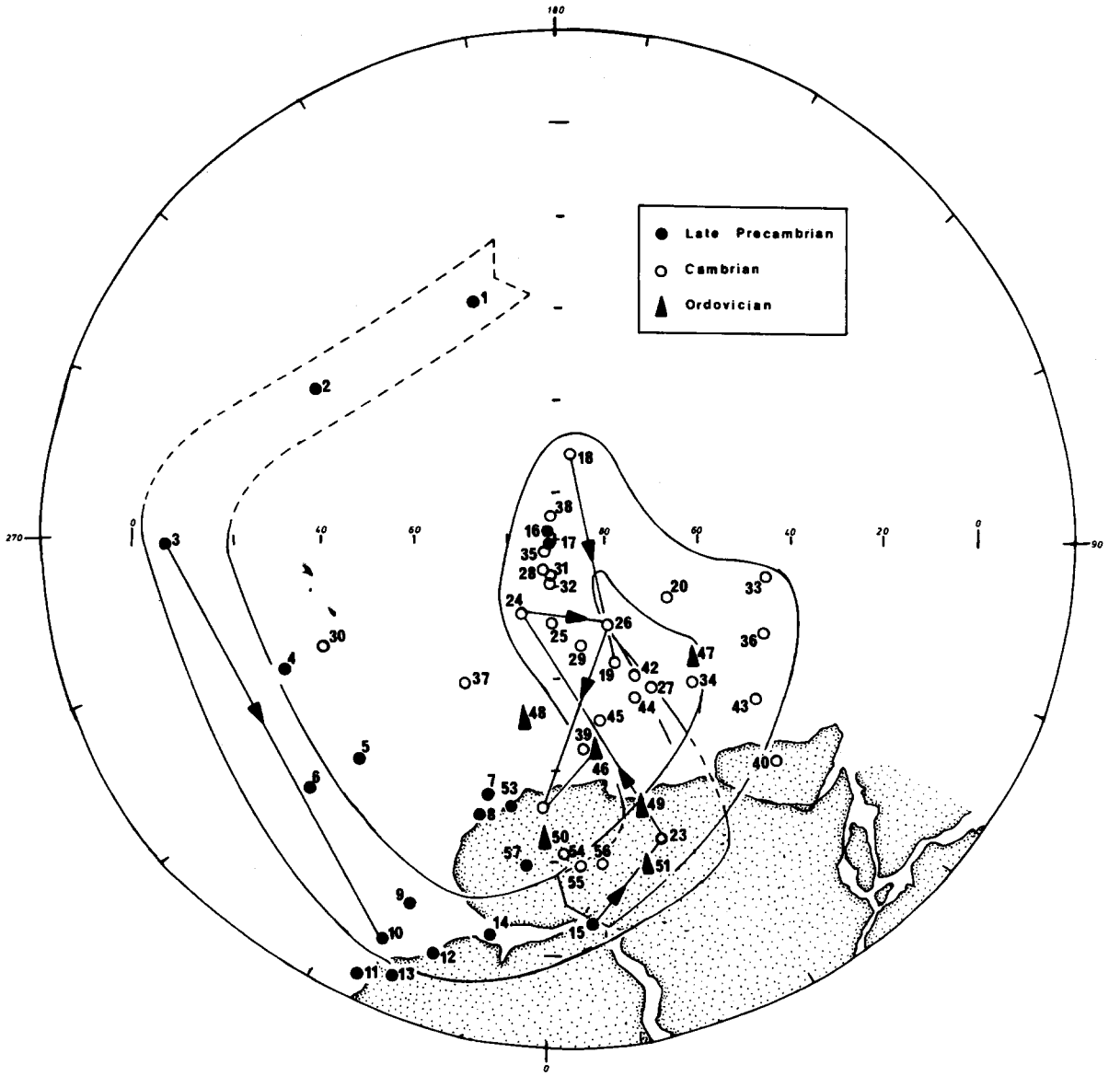


Fig. 8. The distribution of palaeomagnetic poles from Gondwanaland assigned to the interval ca. 650 Ma to Silurian times (Table IV) together with a generalised a.p.w. path. Critical time sequences of poles which assist in the definition of this path are indicated by the arrowed lines.

a.p.w. path (Kröner et al., 1980) with the sequence of magnetisations N2 → N3 → Nama secondary, and it is possible that this path should also be extended back to the stratigraphically-lowest (N1) magnetisation yielding a palaeopole near the present pole and incorporate pole 16 in Table IV; this has not been included in Fig. 8 because it lies in the proximity of poles of Late Precambrian or Early Cambrian age. As is the case with the Laurentian shield, there are only a few data points from Gondwanaland relevant to the interval 800–650 Ma, and with the suggestion that a long a.p.w.

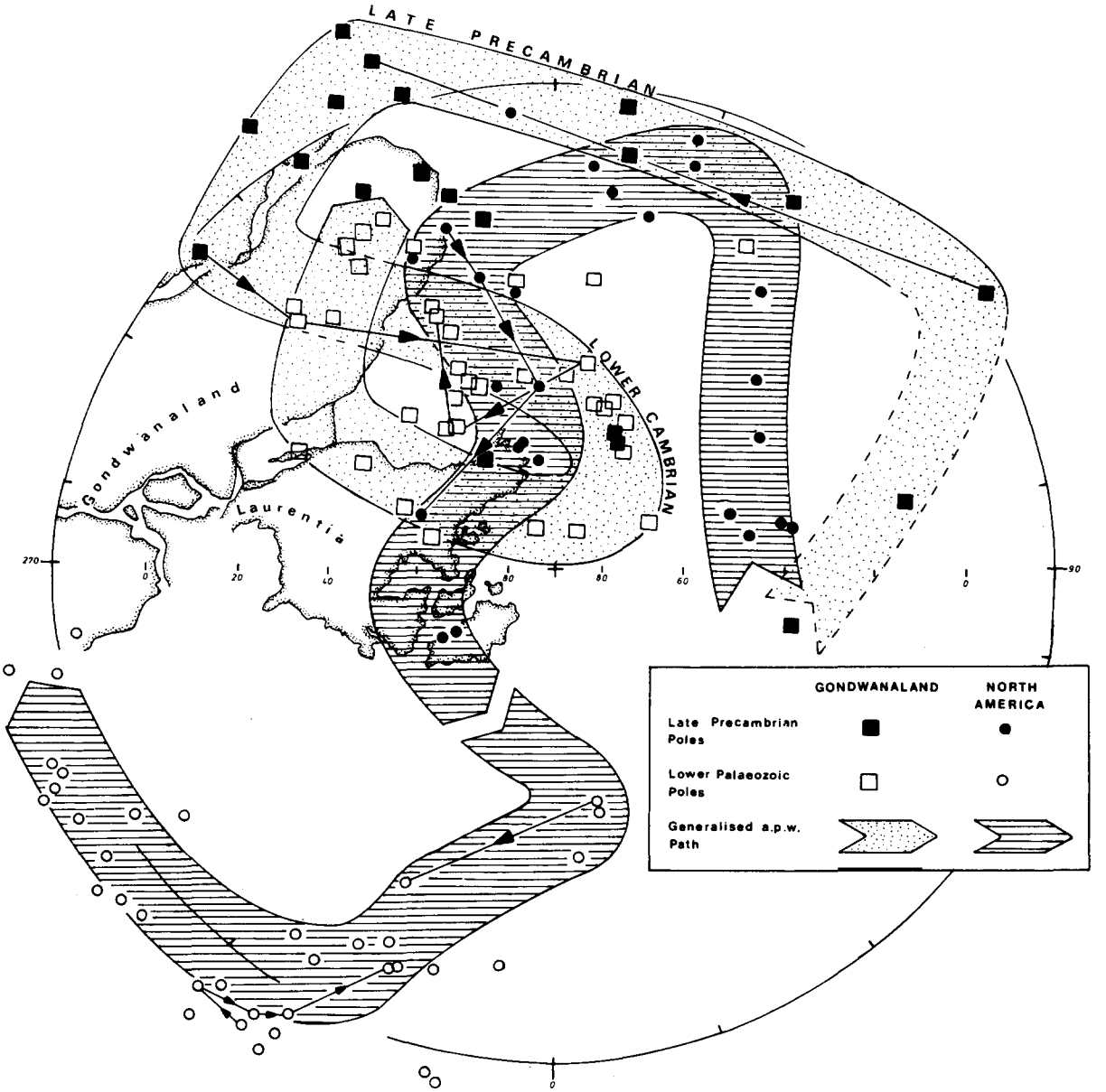


Fig. 9. The palaeopoles from Gondwanaland (squares) of Fig. 8 and Table IV rotated according to the Proterozoic supercontinent reconstruction for comparison with the Laurentian poles (circles) of Fig. 7 and Table III. The subdivision into poles of Late Precambrian (closed symbols) and Early Cambrian (open symbols) is not rigid in the case of the Laurentian data, and the a.p.w. paths are generalised (see text). Note the clear divergence of the a.p.w. paths in Early Cambrian times.

movement is present prior to 630 Ma (Fig. 8 and Kröner et al., 1980) it is premature to suggest a complete a.p.w. path for this time period; indeed, it is possible that Gondwanaland was not an integral unit in the conventional form (Smith and Hallam, 1970) at this time (McElhinny and Embleton, 1976).

Prior to about 800 Ma there is a close corre-

spondence between the African and Laurentian a.p.w. paths over the whole Precambrian record (Piper, 1976, 1980) producing a contiguous reconstruction involving a clockwise rotation of Africa by 146° about a Euler pole at 73°N 138°E. The reconstruction is substantiated by a wide range of geological evidence (Piper, 1976), and whilst there is no doubt that the continental crust was subject to widespread internal deformation, a large part at least retained a coherent entity during progressive cooling (Piper, 1980). The correspondence of the ca. 1050–850 Ma a.p.w. loops from the major shields shows however that the Fennoscandian Shield broke up and rotated with respect to the other shields following the Grenville–Sveconorwegian mobile episode (Patchett et al., 1978; Piper, 1980); also the relationship of 'Eastern Gondwanaland' (the a.p.w. path of which does not match the path from Africa/South America on the Conventional reconstruction (McElhinny and Embleton, 1976)) has yet to be clarified.

As a first step in the analysis of Late Precambrian–Cambrian continental configuration it is logical to examine the Africa/South America Laurentian shield reconstruction derived from the pre-800 Ma data to establish when this reconstruction ceased to be applicable, and hence determine when this largest segment of the Proterozoic supercontinent was dismembered. In Fig. 9 the palaeomagnetic poles of Table III and Fig. 8 from Gondwanaland are rotated clockwise by 146° about the Euler pole at 73°N 138°E for comparison with the Hadrynian Track of Fig. 7. When this operation is completed there is seen to be a close correspondence between these two south a.p.w. paths prior to Early Cambrian times. Pre-600 Ma anticlockwise loops are present in both paths although they can only be used as ancilliary evidence because their age control is so poor. However the sequence of ca. 580–560 Ma pole positions from West Greenland defines a similar motion to the stratigraphic sequence of Late Precambrian to Early Cambrian poles from Australia (Fig. 9); the probable sequence of magnetisations from the Lamotte Formation (Fig. 7 and McElhinny et al., 1974) would also conform to the Gondwanaland path.

The generalised a.p.w. swathes in Fig. 9 are drawn by taking the data in Tables III and IV at their face value. There are several points which suggest that the correspondence between the paths may be even closer than indicated here; these are:

(1) The 'Late Precambrian' poles which produce the somewhat deeper lobe in the Gondwanaland a.p.w. path compared with the Laurentian path at ca. 600 Ma are from peripheral parts (Australia and India) of Gondwanaland for which the reconstruction (at this time) is less certain specifically because of possible deformation along the ca. 650–450 pan-African belts. Pole 11 from Newfoundland suggests that this lobe should possibly be slightly deeper in the Laurentian path.

(2) The lobe is extended on the Gondwanaland path by the pole from the Pound Quartzite, although it is now believed (McWilliams and McElhinny, 1980) that this may represent a Silurian secondary magnetisation. The new results of Klootwijk (1980, poles 33–57) however constrain the age of the loop defined by McElhinny et al. (1974) between latest Precambrian and earliest Cambrian times (poles 53 and 57) and late Middle-Cambrian times (poles 55 and 56). These latter results also increase the rate of implied a.p.w. motion at this time and make it unlikely that the contemporary Laurentian data record the full a.p.w. path: the important point in Fig. 9 is the exact correspondence of the post-basal Cambrian tracks.

(3) The Greenland poles and secondary magnetisations defining the Hadrynian Track are indicated as Latest Precambrian (Fig. 9, closed symbols) although they could equally well be Early Cambrian in age in conformity with most of the Gondwanaland poles of this established stratigraphic age and with which they are identical.

The reconstruction of Fig. 9 shows quite clearly that this largest part of the Proterozoic supercontinent did not break up until Early Cambrian times because this is the same unique reconstruction as that applicable to the earlier Proterozoic palaeomagnetic poles. The paths of Fig. 9 should then be extrapolated back to the matching 1050–850 Ma a.p.w. loops (Piper, 1980) and they define the conventional a.p.w. paths for the pre-800 Ma

data illustrated in the Pacific areas and western Atlantic (McElhinny et al., 1974; Irving and McGlynn, 1976; Piper, 1976) probably as south a.p.w. paths. A comparison of the defined sequences of magnetisation in Fig. 9 would time the continental break-up between poles 24 and 26 (the Lower Cambrian Hudson Formation and the Lower–Middle Cambrian High River Shale); beyond this the a.p.w. paths are quite clearly divergent. This finding contributes towards the explanation of a range of geological observations some of which are outlined here.

(1) A marine regression followed by a transgression was a worldwide phenomenon in Lower Cambrian times: it is now clear that this was associated with continental break-up and generation of a new ocean basin.

(2) The kimberlite–lamprophyre suite in Greenland is part of a widespread alkaline province dated between 600 and 554 Ma and extending from the Superior Province through Greenland to the Fennoscandian shield and probably also including the Siberian shield (Doig, 1970); it is spatially-associated with profound graben faulting and reactivation of ancient basement structures. Figure 9 shows that this magmatic tectonic episode immediately predated the breakup of the continental crust. Mesozoic kimberlite emplacement also began with continental break-up of Africa and South America (Dawson, 1970) and continued for several tens of millions of years during the initial opening of the South Atlantic.

(3) The Iapetus Ocean (Harland and Gayer, 1972) is associated with early tectonic phases which culminated in the Caledonian orogenic belt along the southeast margin of the Laurentian shield. The south-easterly provenance of Late Precambrian/earliest Cambrian rocks, notably the Portaskaig tillite, implies that this ocean did not open until Cambrian times (Anderton, 1980); the palaeontological evidence, however, requires that distinct faunal provinces were established in Lower Cambrian times (Cowie, 1971). These observations are in conformity with the timing of relative movements suggested here. The Iapetus Ocean was ultimately closed by the re-impingement of Gondwanaland and the Laurentian shield in late Ordovician or Silurian times (McKerrow and Ziegler, 1972; Piper, 1979).

(4) The post-Early Cambrian relative movements of the Laurentian and Gondwanaland Shields implied by the divergent a.p.w. paths of Fig. 9 necessitate little modification of existing palaeogeographic maps. However, palaeomagnetism cannot closely constrain the maximum separation achieved prior to convergence because the Laurentian shield lay in low latitudes after Early Cambrian times (Fig. 7).

(5) The reconstruction of Fig. 9 suggests that the Late Precambrian to Early Cambrian tillite horizons were not a synchronous world event because West Africa lay in high latitudes while Greenland lay in low latitudes, although the time interval between them may not be more than 20–30 Ma. The low latitude sediments between the Late Precambrian and Lower Palaeozoic tillites in West Africa are most likely to be the broad equivalent of the tillite horizon in Greenland, northern Norway and Scotland. Unfortunately, Rb–Sr and palaeomagnetic studies are unlikely to produce a definitive answer to this question because of the difficulty of assigning a depositional age to either the isotopic or the palaeomagnetic data (see Morris, 1977; and McWilliams and McElhinny, 1980 for recent discussions).

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